

Assignment 5 - Bug Fixing & Code Log

File	Line	Classification	Error Identified	Resolution Patch
1.c	14	Recursion Error	Missing base case in recursive function, causing infinite recursion and Stack Overflow.	Add a base condition at the beginning: <code>if (n <= 1) return 1;</code>
2.c	27	Matrix Boundary	Using incorrect row/column index limits (swapping rows and columns loops) while traversing a 2D array.	Correct the nested loop bounds: <code>for(int i=0; i<ROWS; i++)</code>
3.c	45	File Handle Leak	Opening a file streams resource via <code>fopen()</code> but forgetting to close it using <code>fclose()</code> before function exit.	Add close call before returning: <code>fclose(filePtr);</code>
4.c	19	Bitwise Mistake	Using logical OR (<code> </code>) instead of bitwise OR (<code> </code>) for bit masking operations.	Change logical operator to bitwise: <code>flags = flags MASK;</code>
5.c	33	Type Mismatch	Passing a structure directly to a function that expects a structure pointer reference argument.	Pass by address using ampersand: <code>processData(&currentStruct);</code>